INTELLIGENT WATER BODY MAPPING FROM MULTI-SOURCE SATELLITE OBSERVATIONS USING DEEP LEARNING

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ABOUT THE PROJECT

Accurate water-body detection is essential for environmental monitoring, flood assessment, coastal management, and climate-change studies. However, identifying water from satellite imagery can be challenging due to clouds, shadows, seasonal variations, and the complexity of different landscapes.

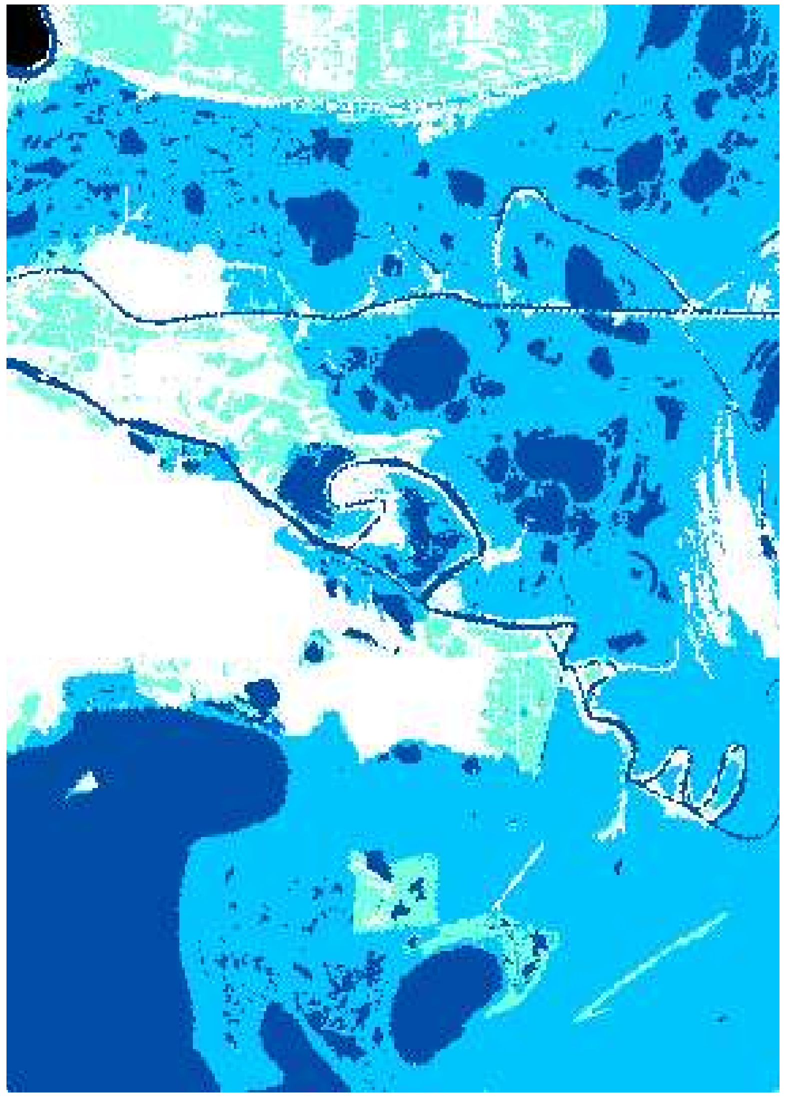
To address this challenge, this project uses a U-Net deep learning model trained on 12-channel multi-source satellite data. The input data combines spectral information from Sentinel-2/Landsat imagery with elevation models, land-cover maps, quality assessment layers, and historical water occurrence information.

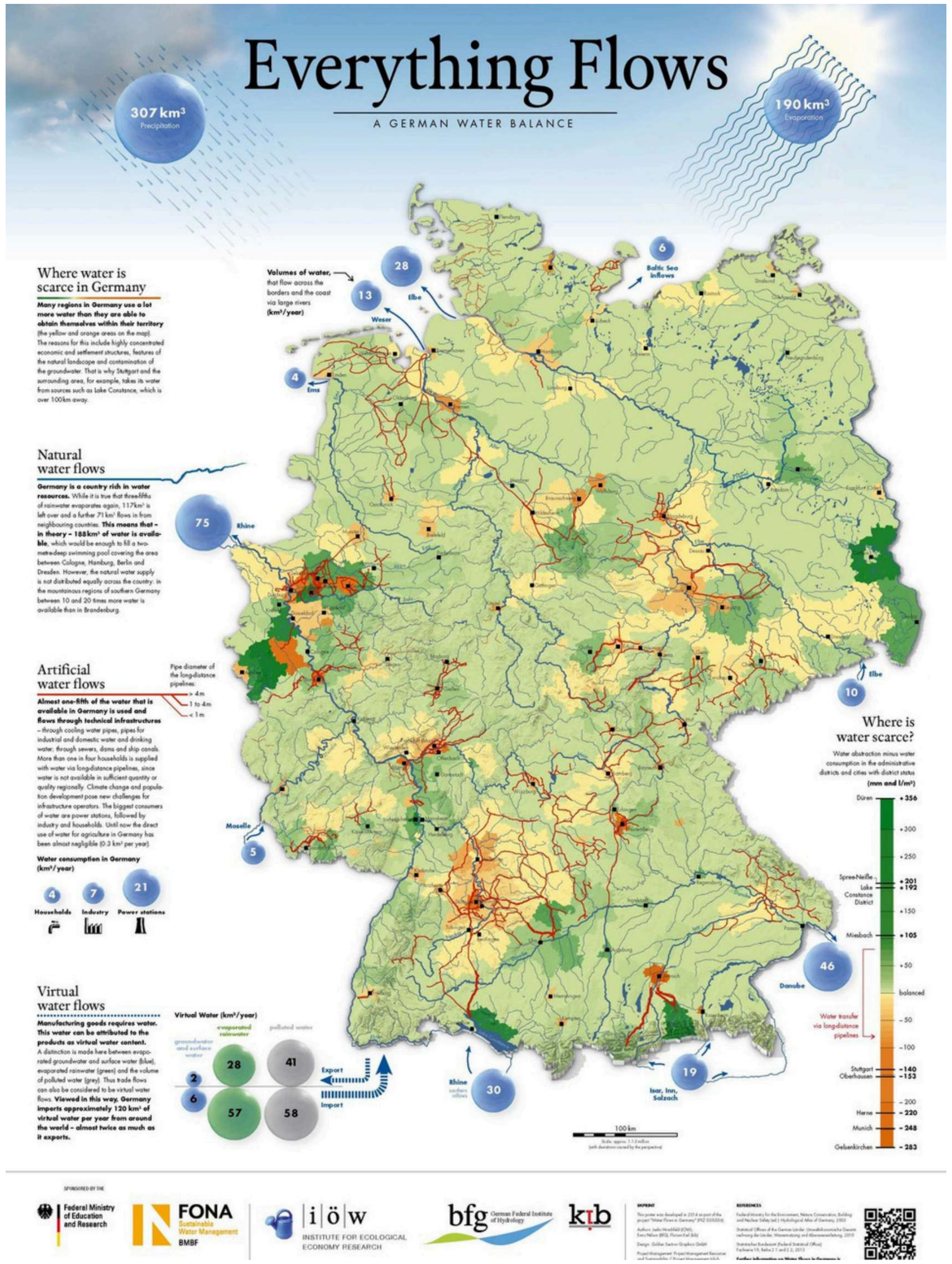
By integrating these complementary data sources, the model can better distinguish water from non-water regions and automatically generate water masks, providing a scalable solution for water-resource monitoring and ocean science applications.

WHY IS WATER MAPPING IMPORTANT?

Water bodies continuously change due to:

- Climate change
- Flooding
- Coastal erosion
- Sea level rise
- Human activities





"Water is one of the most dynamic components of the Earth system. Monitoring its spatial distribution is essential for understanding environmental change, coastal processes, and aquatic ecosystems."

PROJECT OBJECTIVE

Develop an AI model capable of:

- Detecting water bodies
- Producing accurate water masks
- Supporting ocean and environmental monitoring

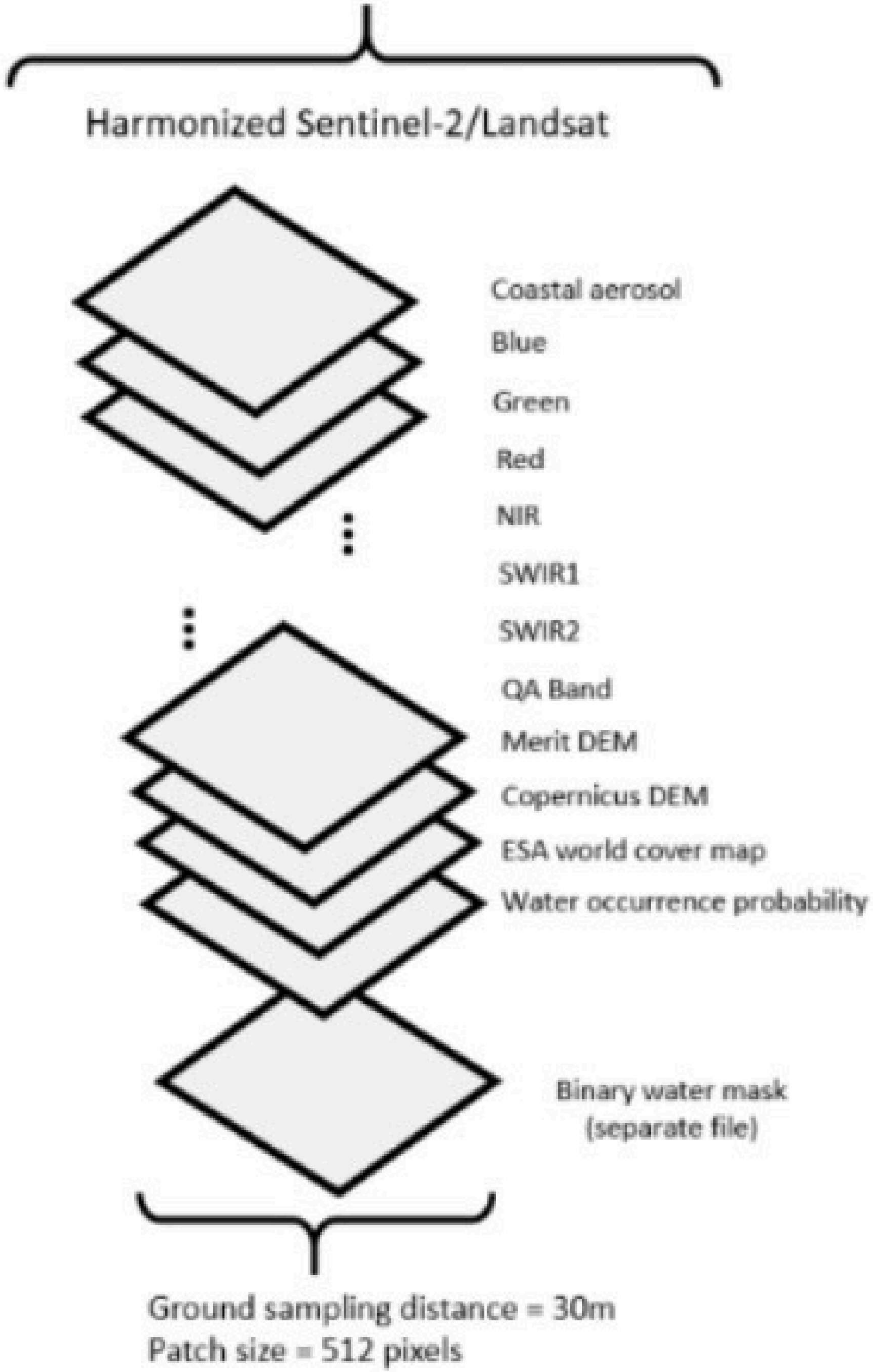
Can a U-Net deep learning model accurately segment water bodies using multi-source satellite imagery and geospatial datasets?

Connection to Ocean Science

This project can support:

- Coastal monitoring
- Estuary mapping
- Wetland assessment
- Flood detection
- Tracking shoreline changes
- Climate impact studies

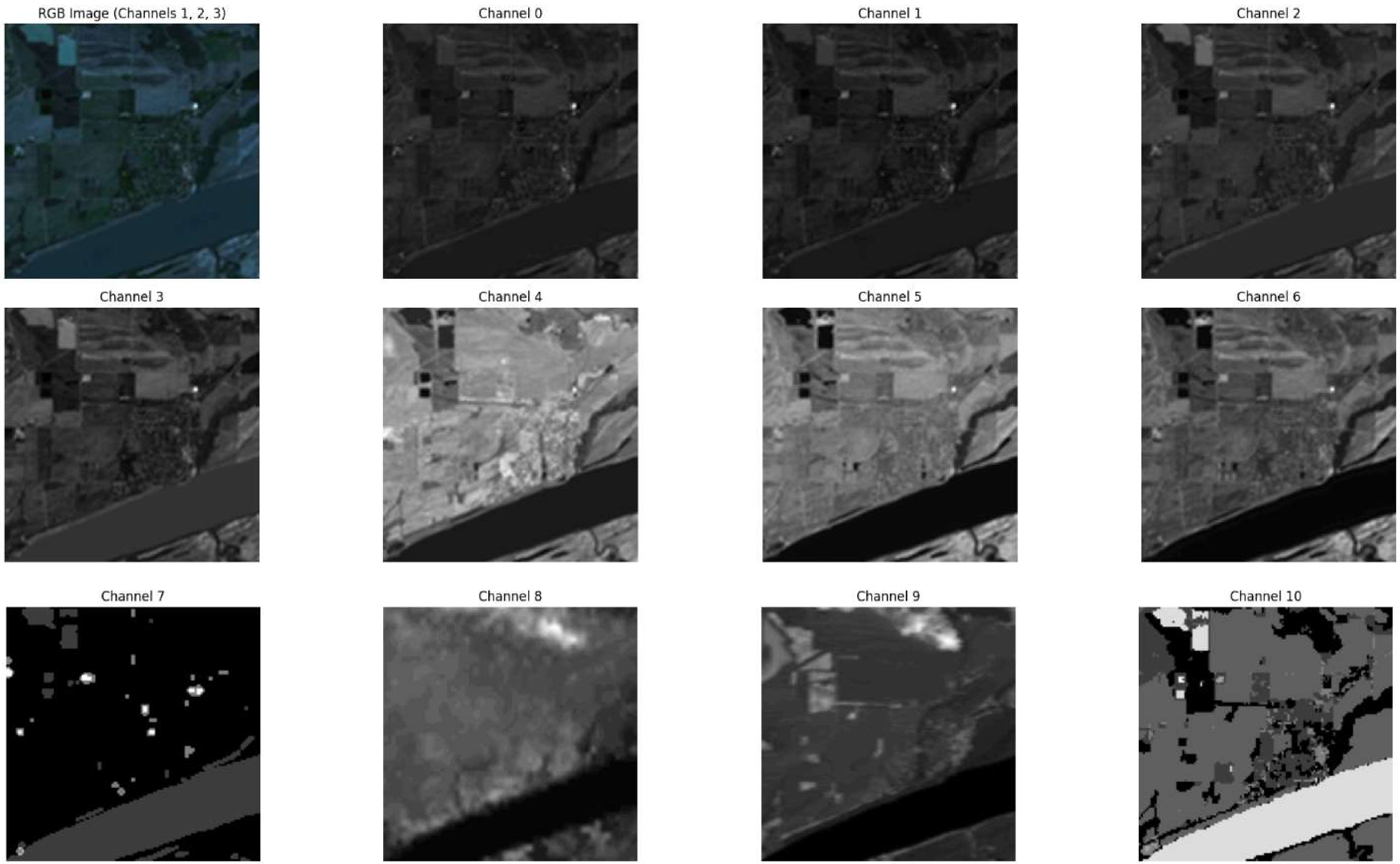
OCEAN SCIENCE RELEVANCE

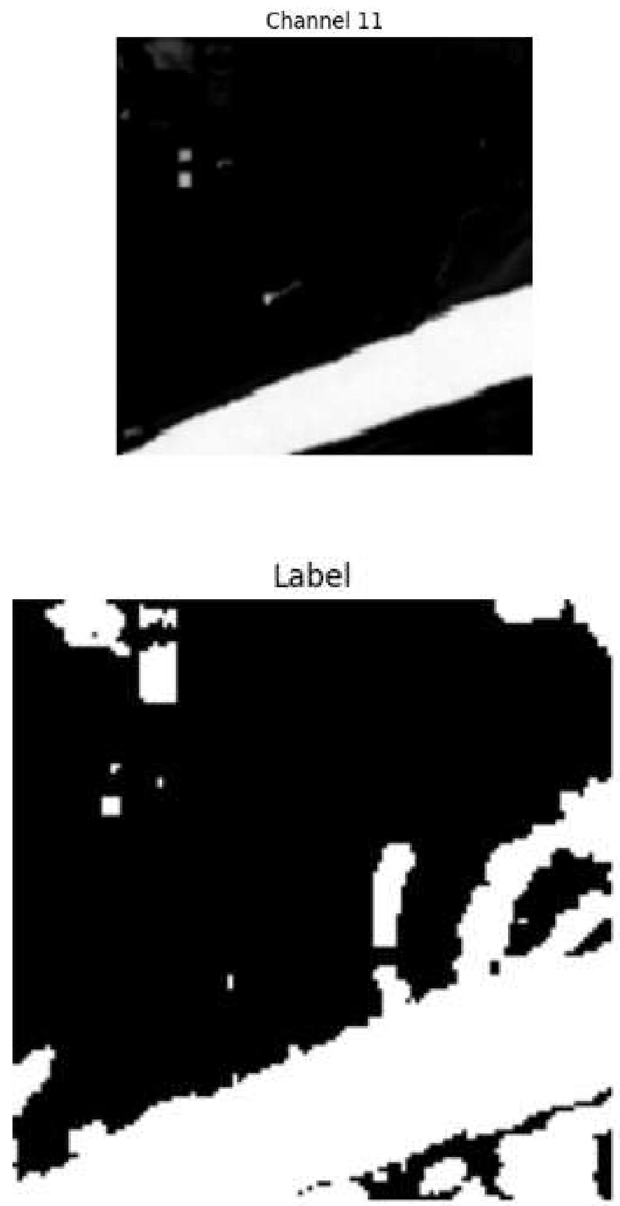


DATASET OVERVIEW

The model uses:

Band	Purpose	Layer	Purpose
Coastal Aerosol	Coastal water observation	SWIR2	Surface characterization
Blue	Water penetration	QA Band	Cloud quality control
Green	Vegetation-water contrast	MERIT DEM	Elevation information
Red	Surface reflectance	Copernicus DEM	Terrain information
NIR	Water strongly absorbs NIR	ESA World Cover	Land cover classification
SWIR1	Moisture detection	Water Occurrence Probability	Historical water presence



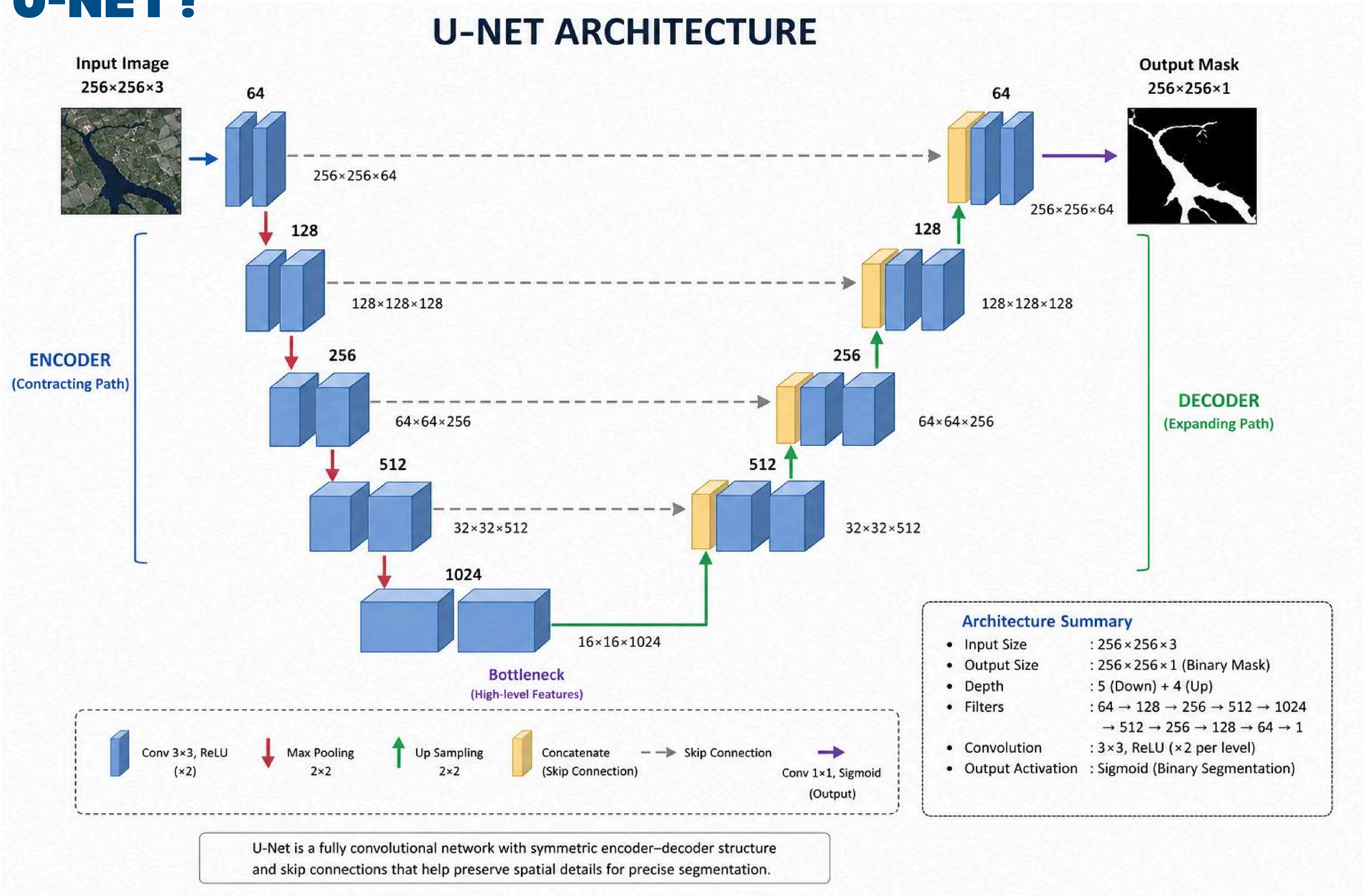


Limited dataset size (305 samples)

Patch Size: 512 × 512 pixels Ground Sampling Distance: 30 m Meaning: Each pixel represents 30 meters on the Earth'ssurface.

Data < Link >

WHY U-NET?



U-NET ARCHITECTURE
Input Image
256×256×3
64
256×256×64
128
128×128×128
256
64×64×256
512
32×32×512
1024
16×16×1024
Bottleneck
(High-level Features)
ENCODER
(Contracting Path)
DECODER
(Expanding Path)
Output Mask
256×256×1
Conv 3×3, ReLU (×2)
Max Pooling 2×2
Up Sampling 2×2
Concatenate (Skip Connection)
Skip Connection
Conv 1×1, Sigmoid (Output)
Architecture Summary
Input Size : 256×256×3
Output Size : 256×256×1 (Binary Mask)
Depth : 5 (Down) + 4 (Up)
Filters : 64 → 128 → 256 → 512 → 1024 → 512 → 256 → 128 → 64 → 1
Convolution : 3×3, ReLU (×2 per level)
Output Activation : Sigmoid (Binary Segmentation)
U-Net is a fully convolutional network with symmetric encoder–decoder structure and skip connections that help preserve spatial details for precise segmentation.

U-NET ARCHITECTURE

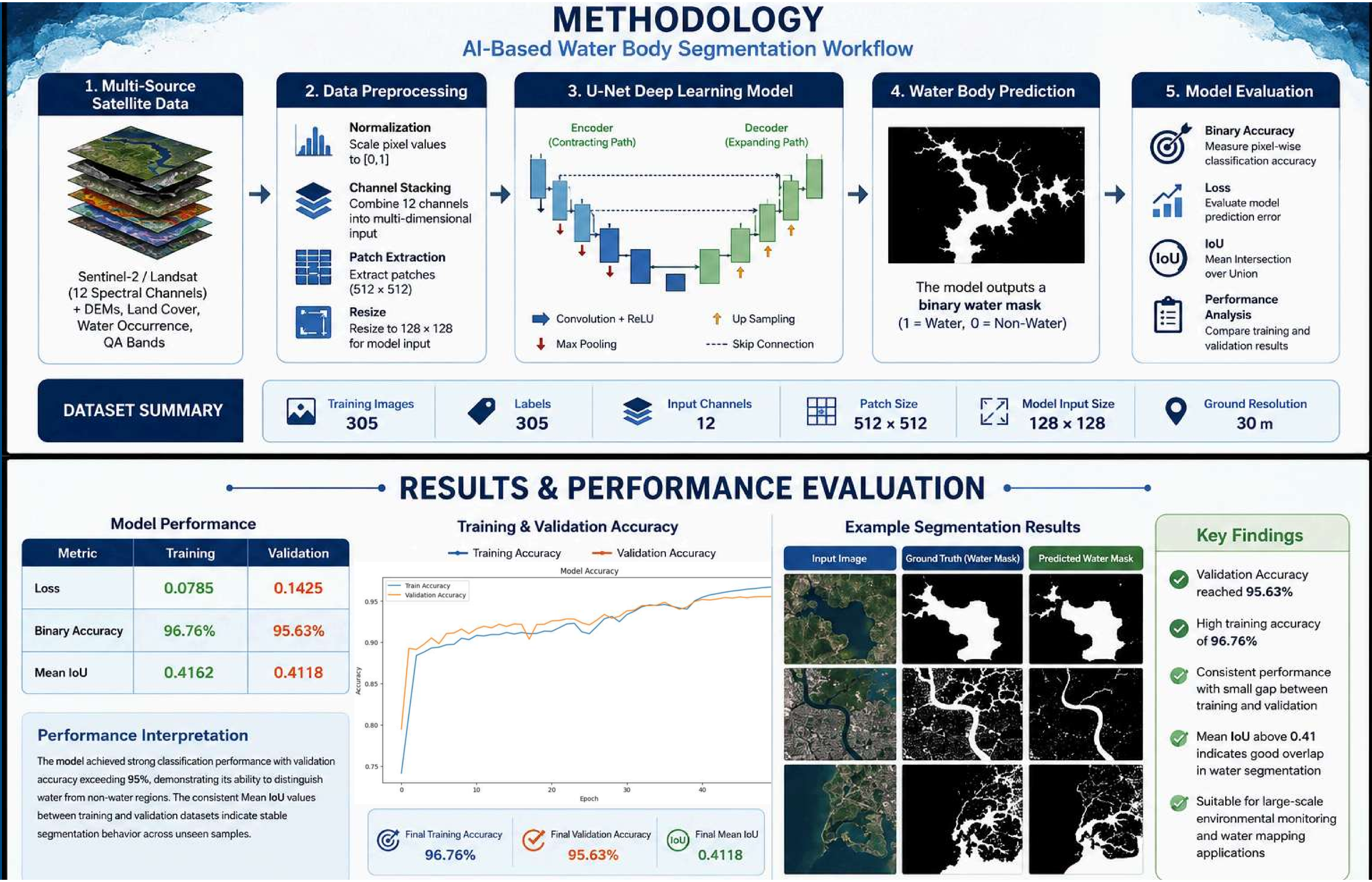
Originally developed for image segmentation.

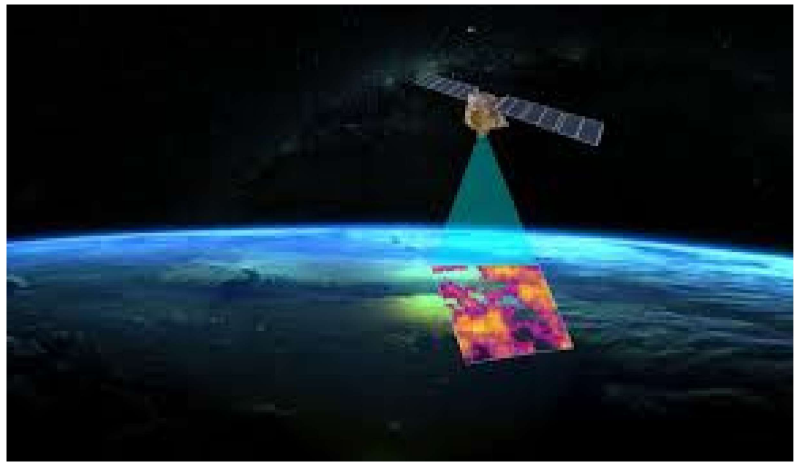
Very effective because:

- **Learns spatial patterns**
- **Preserves fine boundaries**
- **Works well with limited training data**
- **Produces pixel-level classification**

U-Net is one of the most widely used architectures for segmentation tasks.

Input Satellite Image → Predicted Water Mask


METHODOLOGY
AI-Based Water Body Segmentation Workflow
1. Multi-Source Satellite Data
Sentinel-2 / Landsat (12 Spectral Channels) + DEMs, Land Cover, Water Occurrence, QA Bands
2. Data Preprocessing
Normalization
Scale pixel values to [0,1]
Channel Stacking
Combine 12 channels into multi-dimensional input
Patch Extraction
Extract patches (512 × 512)
Resize
Resize to 128 × 128 for model input
3. U-Net Deep Learning Model
Encoder (Contracting Path)
Decoder (Expanding Path)
Convolution + ReLU
Up Sampling
Max Pooling
Skip Connection
4. Water Body Prediction
The model outputs a binary water mask (1 = Water, 0 = Non-Water)
5. Model Evaluation
Binary Accuracy
Measure pixel-wise classification accuracy
Loss
Evaluate model prediction error
IoU
Mean Intersection over Union
Performance Analysis
Compare training and validation results
DATASET SUMMARY
Training Images 305
Labels 305
Input Channels 12
Patch Size 512 × 512
Model Input Size 128 × 128
Ground Resolution 30 m
RESULTS & PERFORMANCE EVALUATION
Model Performance
Metric | Training | Validation
Loss | 0.0785 | 0.1425
Binary Accuracy | 96.76% | 95.63%
Mean IoU | 0.4162 | 0.4118
Performance Interpretation
The model achieved strong classification performance with validation accuracy exceeding 95%, demonstrating its ability to distinguish water from non-water regions. The consistent Mean IoU values between training and validation datasets indicate stable segmentation behavior across unseen samples.
Training & Validation Accuracy
Training Accuracy
Validation Accuracy
Model Accuracy
Train Accuracy
Validation Accuracy
Accuracy
Epoch
Final Training Accuracy 96.76%
Final Validation Accuracy 95.63%
Final Mean IoU 0.4118
Example Segmentation Results
Input Image
Ground Truth (Water Mask)
Predicted Water Mask
Key Findings
Validation Accuracy reached 95.63%
High training accuracy of 96.76%
Consistent performance with small gap between training and validation
Mean IoU above 0.41 indicates good overlap in water segmentation
Suitable for large-scale environmental monitoring and water mapping applications



Why This Approach Is Powerful

✓ Fully automated
✓ Scalable globally
✓ Works on large satellite datasets
✓ Reduces manual mapping
✓ Useful for near real-time monitoring

ADVANTAGES

Challenges

- Clouds
- Shadows
- Mixed pixels
- Optimized for 12-channel satellite data
- Seasonal variation
- Small water bodies
- Limited dataset size
- Reduced accuracy on RGB-only images

Potential improvements:

- Attention U-Net
- DeepLabV3+
- Transformer-based segmentation
- Dedicated RGB-compatible version

LIMITATIONS

TRY MY MODEL :)

1-Select Satalite 12-channels Image :

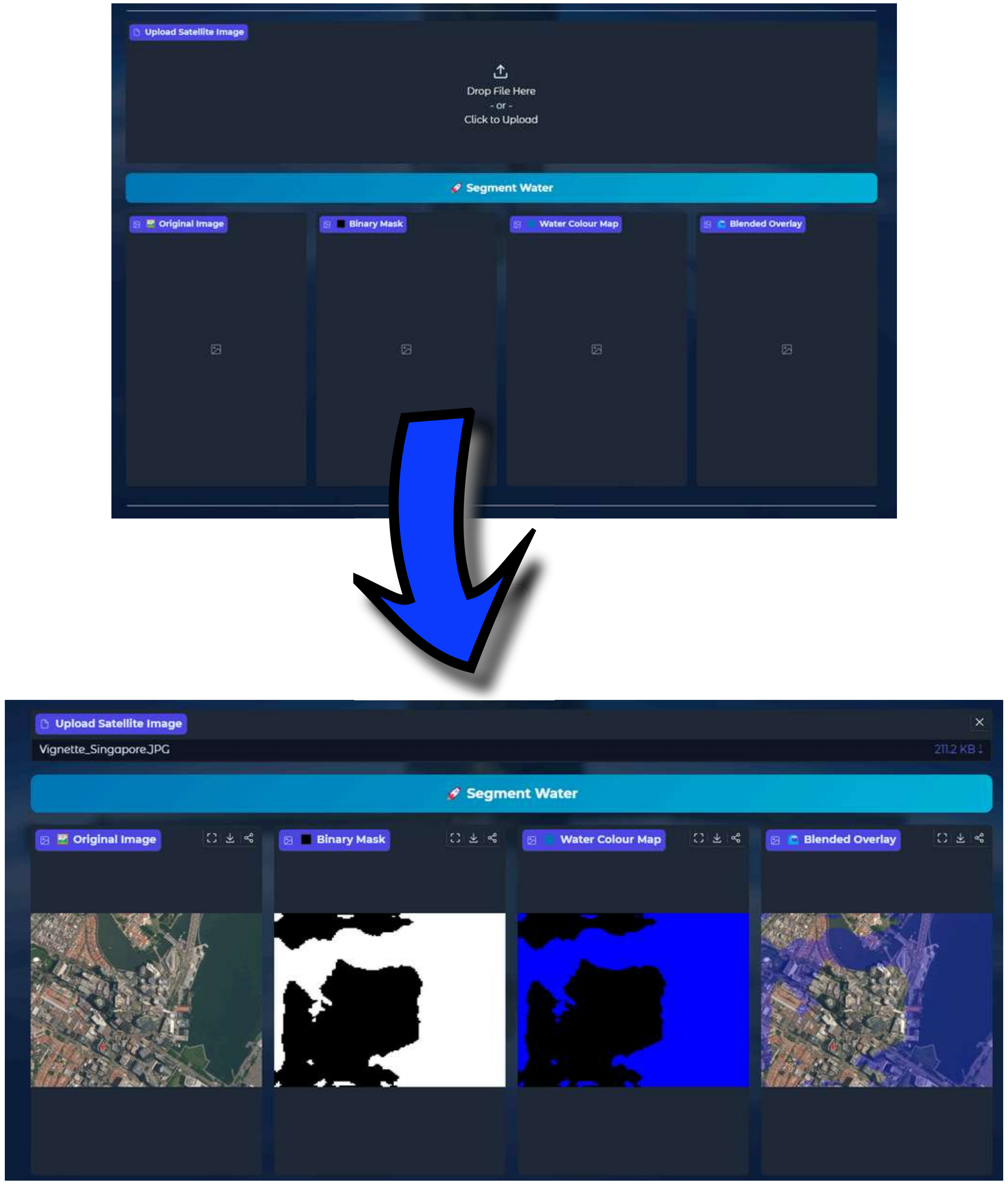
(Here)

2-go through this link and upload the Image :

(Live Demo)

Und... lauft!

Repo Link



ABOUT ME

Master’s student at rostock universitat specializing in Ocean Engineering and Renewable Energy Technologies, with a foundation in Mechatronics and Ai . Interested in renewable energy technologies, wind energy, digitalization, and AI-driven solutions that accelerate the transition to sustainable energy.

DATA SCIENTIST — CodSoft (Code.Create.Succeed)

COMPUTER VISION — Cellula Technologies

NLP ENGINEER — Cellula Technologies

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INTERSHIPS

Thank you for your time.

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